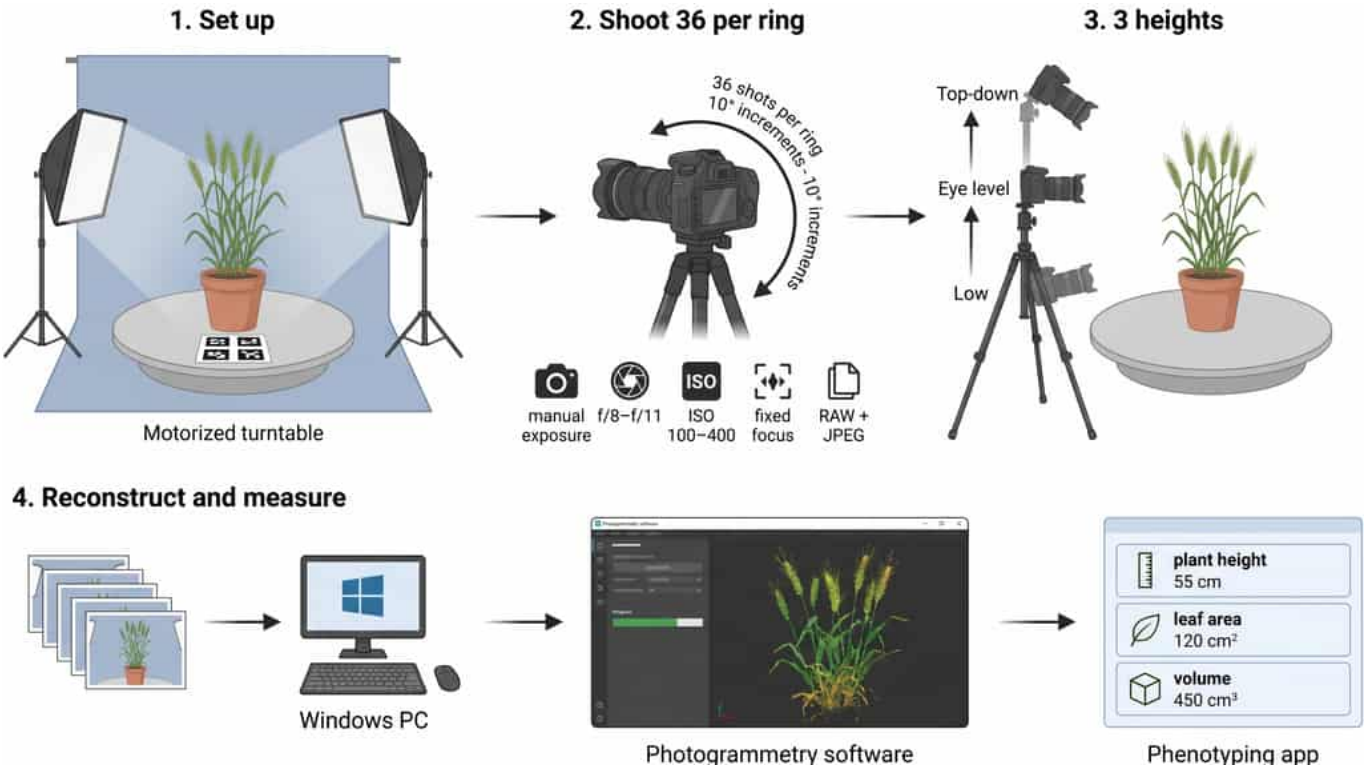
SOP-3 DSLR / mirrorless camera

High-detail multi-view photogrammetry on a turntable

SOP-3 · PhenoForge capture SOP · version 1.0 (October 2026)



What you need

Checklist

- DSLR or mirrorless body, 35–50 mm (full-frame equivalent) prime lens
- Tripod, remote release or interval timer
- Turntable (manual with 10° marks, or motorised)
- Two softboxes / LED panels, plain grey or blue backdrop (not green)
- ArUco card on the turntable
- Colour chart for the first shot of each session

Camera settings

Setting	Value	Why
Mode	Manual (M)	Identical exposure in every photo
Aperture	f/8 – f/11	Depth of field covers the whole plant
ISO	100 – 400	Low noise keeps fine leaf texture
Shutter	≥ 1/60 s on a tripod (use remote)	No camera shake
Focus	Manual, set once on the plant centre	Focus changes the lens geometry
Stabilisation	Off on a tripod	Prevents tiny image shifts
Format	RAW + JPEG	RAW for colour work; JPEG is enough for 3D
White balance	Fixed (daylight or custom)	Comparable colour between sessions

Capture

1. **Set up**
   Plant centred on the turntable, markers flat on the turntable, lights at 45° on both sides, backdrop 1 m behind.
2. **Shoot ring 1 (low)**
   Camera at pot height, plant filling ~60 % of the frame. Rotate 10° between shots: 36 photos.
3. **Rings 2 and 3**
   Raise the tripod to eye level (≈30° down) and to near top-down (≈60° down); 36 photos each.
4. **Copy to the PC**
   Copy the whole card folder (RAW and JPEG) into one folder per plant, named with the plant ID and date.

In PhenoForge

1. **Import**
   Tools > Import from a capture device, device *DSLR / mirrorless camera*, **Photo folder…**. RAW files are developed with the camera white balance (JPEGs with the same name are skipped), blurred shots are rejected, and an optional colour correction from the Colour tab is applied.
2. **Reconstruct and scale**
   COLMAP builds the dense cloud, which opens in the 3D analysis tab. Choose **ArUco markers** and type the marker size.
3. **Segment**
   Section 3: *single* for one plant, *watershed* for a tray of touching plants (enter the number of plants), or *grid* with rows, columns and pot pitch for a bench of equally spaced pots.
4. **Run and export**
   Press **Run analysis**. Results appear in the Scan / Per plant / Leaves tables; **Save results** writes JSON and CSV. Colour the view by Height, Greenness or Plants to check the segmentation.

Tips

- A turntable moves the plant, so the background must be plain: a textured background that does not rotate confuses the reconstruction.
- Leaves that move as the turntable starts and stops cause doubling: wait 2 s after each rotation.
- 108 photos at 24 MP take about 20–60 min in COLMAP on a desktop GPU.